Reminder: Inverse Functions

Defintion A function f is *invertible* if the following equivalent conditions are fulfilled:

- f is a one-to-one function;
- $x \neq y \Rightarrow f(x) \neq f(y)$;
- $f(x)$ passes the horizontal line test.

If $f^{-1}(x)$ is the inverse of f, then

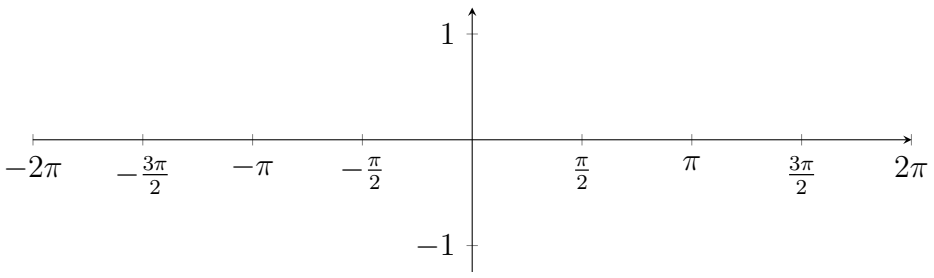
- $f(f^{-1}(x)) = x$ for all x in the domain of f^{-1}, and $f^{-1}(f(x)) = x$ for all x in the domain of f.
- $y = f^{-1}(x) \Leftrightarrow x = f(y)$.
- $\text{Domain}(f^{-1}) = \text{Range}(f)$ and $\text{Range}(f^{-1}) = \text{Domain}(f)$.
- The graphs of $f(x)$ and $f^{-1}(x)$ are symmetric about the line $y = x$.

Inverting Trigonometric Functions

Now let's apply what we know about inverse functions in general to the specific functions $y = \sin x$ and $y = \tan x$.

sin x

On the axes below graph $y = \sin x$ on $[-2\pi, 2\pi]$.



Why is $\sin x$ not invertible on $[-2\pi, 2\pi]$?

What is the simplest domain on which $\sin x$ is invertible?

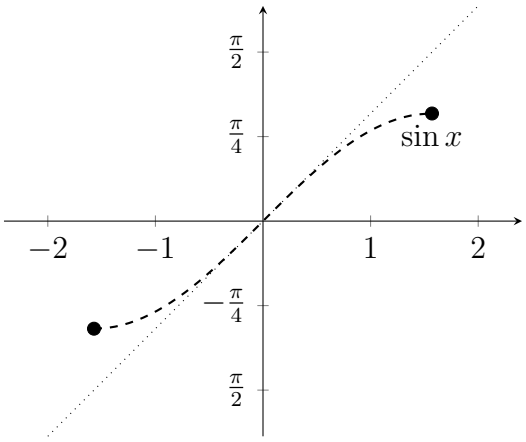
Let's call the inverse of $\sin x$ on that domain $\sin^{-1} x$, or $\arcsin x$. What is the domain of $\sin^{-1} x$? The range?

What is $\sin(\sin^{-1} x)$? For which values of x is that true?

What is $\sin^{-1}(\sin x)$? For which values of x is that true?

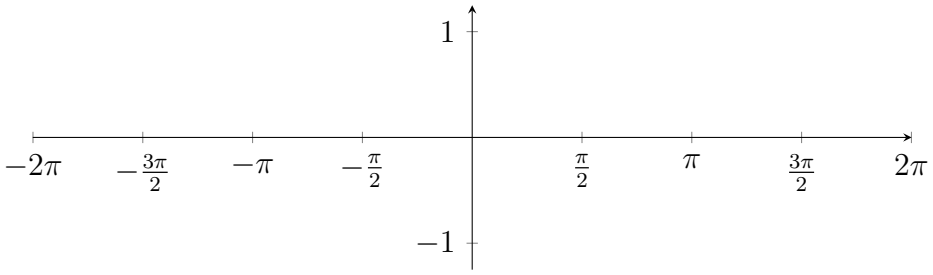
If $y = \sin^{-1} x$, then $x =$ _____.

The axes below show a graph of $\sin x$ on the domain $[-\frac{\pi}{2}, -\frac{\pi}{2}]]$. Sketch a graph of $y = \sin^{-1} x$. The line $y = x$ is shown dotted to help you.



COS x

On the axes below graph $y = \cos x$ on $[-2\pi, 2\pi]$.



Why is $\cos x$ not invertible on $[-2\pi, 2\pi]$?

What is the simplest domain on which $\cos x$ is invertible?

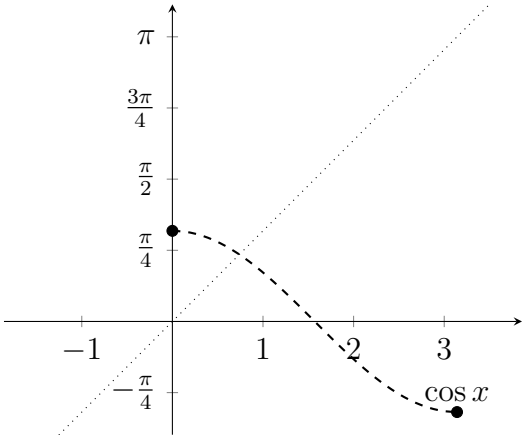
Let's call the inverse of $\cos x$ on that domain $\cos^{-1} x$, or $\arccos x$. What is the domain of $\cos^{-1} x$? The range?

What is $\cos(\cos^{-1} x)$? For which values of x is that true?

What is $\cos^{-1}(\cos x)$? For which values of x is that true?

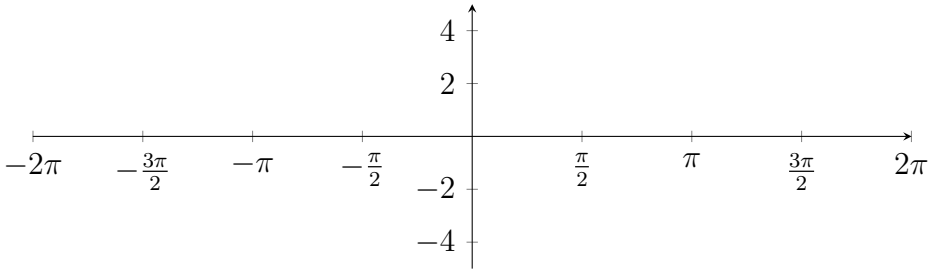
If $y = \cos^{-1} x$, then $x =$ _____.

The axes below show a graph of $\cos x$ on the domain $[0, \pi]$. Sketch a graph of $y = \cos^{-1} x$. The line $y = x$ is shown dotted to help you.



tan x

On the axes below graph $y = \tan x$ on $[-2\pi, 2\pi]$.



Why is $\tan x$ not invertible on $[-2\pi, 2\pi]$?

What is the simplest domain on which $\tan x$ is invertible?

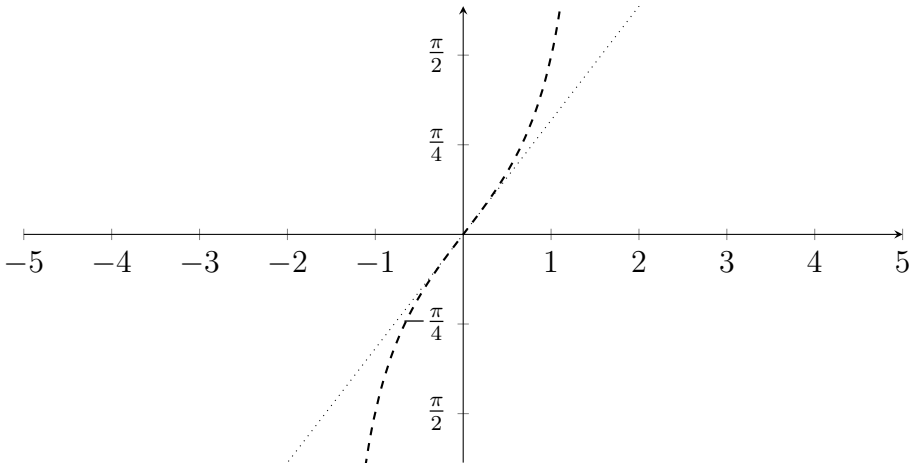
Let's call the inverse of $\tan x$ on that domain $\tan^{-1} x$, or $\arctan x$. What is the domain of $\tan^{-1} x$? The range?

What is $\tan(\tan^{-1} x)$? For which values of x is that true?

What is $\tan^{-1}(\tan x)$? For which values of x is that true?

If $y = \tan^{-1} x$, then $x =$ _____.

The axes below show a graph of $\tan x$ on the domain $(-\frac{\pi}{2}, \frac{\pi}{2})$. Sketch a graph of $y = \tan^{-1} x$. The line $y = x$ is shown dotted to help you. Be sure to indicate any asymptotes!



Homework Questions

1. ☐Regrade (a) For each of the following, complete the blanks. Be sure to express your answers in radians.

i. Since $\sin$ ____ = ____, $\sin^{-1}(0)$ = ____

ii. Since $\sin$ ____ = ____, $\sin^{-1}(1)$ = ____

iii. Since $\sin$ _____ = _____, $\sin^{-1}(-1)$ = _____

iv. Since $\cos$ ____ = ____, $\cos^{-1}(0)$ = ____

v. Since $\cos$ ____ = ____, $\cos^{-1}(1)$ = ____

vi. Since $\cos$ ____ = _____, $\cos^{-1}(-1)$ = ____

vii. Since $\tan$ ____ = ____, $\tan^{-1}(0)$ = ____

viii. Since $\tan$ ____ = ____, $\tan^{-1}(1)$ = ____

ix. Since $\tan$ _____ = _____, $\tan^{-1}(-1)$ = _____

(b) On a unit circle, each of the trig functions takes an _____ as its input.

- $\sin x$ outputs an ____-coordinate.
- $\cos x$ outputs an ____-coordinate.
- $\tan x$ outputs the _____ of the corresponding radius.

(c) Therefore

- $\sin^{-1}(x)$ takes a _____ and returns the corresponding _____.
- $\cos^{-1}(x)$ takes an _____ and returns the corresponding _____.
- $\tan^{-1}(x)$ takes a _____ and returns the corresponding _____.

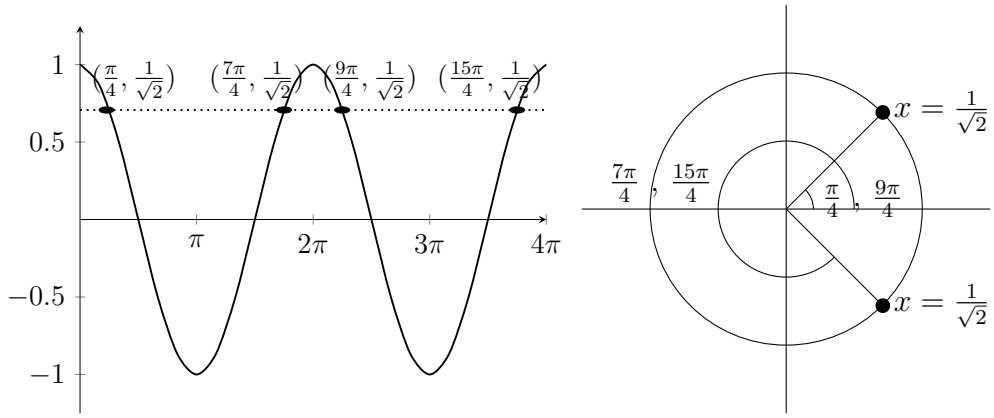
2. For each of the following:

□Regrade

- Draw the given function on the given domain.
- Add in a horizontal line corresponding to the appropriate y-value.
- State how many solutions the equation has.
- Draw a unit circle and mark all points on it that solve the equation with the correct information (see Q1(b) above!).
- Find all the solutions, showing your work. Express your answer in radians, giving exact answers where possible, and expressing your answer to three decimal places where it is not.
- Label angles corresponding to your solutions on your unit circle. Each angle may have more than one label!
- Back on your graph, label all solutions with their coordinates.

The first one is done for you as an example.

(a) $\cos(\theta) = \dfrac{1}{\sqrt{2}}$ for $0 \le \theta \le 4\pi$



There are four solutions.

From left to right: $\cos^{-1}(\frac{1}{\sqrt{2}}) = \frac{\pi}{4}$, $2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$, $2\pi + \frac{\pi}{4} = \frac{9\pi}{4}$, $\frac{7\pi}{4} + 2\pi = \frac{15\pi}{4}$.

(b) $\sin(\theta) = \dfrac{\sqrt{3}}{2}$ for $-\dfrac{\pi}{2} \leq \theta \leq \dfrac{\pi}{2}$

(c) $\tan(\theta) = -\dfrac{1}{\sqrt{3}}$ for $-2\pi \leq \theta \leq 0$

(d) $\cos(\theta) = 0.42$ for $0 \leq \theta \leq 2\pi$

(e) $\sin(\theta) = 0.91$ for $-2\pi \leq \theta \leq 2\pi$

(f) $\tan(\theta) = 2.14$ for $0 \leq \theta \leq 5\pi$

3. For each of the following, solve for all x with $0 \leq x \leq 2\pi$ using the following steps:

□Regrade

- Use algebra to write the equation in the form (e.g.) $\cos x = \frac{1}{2}$. For (e) and (f), you'll need to factor or use the quadratic formula, so you may end up with two such expressions. You should also end up with two in (c).
- Draw a graph of the given function on the domain, and add in lines as you did in Q2. No need to draw unit circles here.
- Find all solutions. Express all your answers in radians and give exact answers if possible. If not, use three decimal places.

(a) $2\cos x = 1$

(b) $\tan x = \sqrt{3} - 2\tan x$

(c) $3\sin^2 x + 4 = 5$

(d) $4\tan x + 3 = 2$

(e) $3\cos^2 x + 2 = 3 - 2\cos x$

(f) $3\sin^2 x + 3\sin x + 4 = 3 - 2\sin x$